

HOMEWORK: MICROFLUIDICS

Microfluidic chip has a channel, which is $L = 20$ cm long, $h = 20$ μ m high, and $w = 50$ μ m wide. Channel is open from both. In one end of the channel there is one electrode (+) and at the distance of $x = 12$ cm along the channel there is another electrode (-). Channel is filled with neutral buffer solution (pH 7) with conductivity 1 S/m. Current 10 μ A is driven through the electrodes.

① Determine the electrical potential between the electrodes and heat dissipated.

Ohm's law: $R = \frac{V}{I}$ $\xrightarrow{\text{What we want to know}}$ electrical potential
 $I \rightarrow$ current $\rightarrow I = 10$ μ A

First we will then have to know the resistance value between the 2 electrodes:

$\sigma = 1$ S/m $\xrightarrow{\text{conductivity}}$ $\rho = \frac{1}{\sigma} = \frac{1}{1 \text{ S/m}} = 1 \Omega \cdot \text{m}$ $\xrightarrow{\text{resistivity}}$

So we're going to have:

$R = \rho \frac{x}{A}$ $\xrightarrow{A \rightarrow \text{channel area}}$ $\Rightarrow R = 1 \cdot \frac{12 \text{ cm}}{(w) \cdot (h)}$
 $w = 50$ μ m $\quad h = 20$ μ m

$\Rightarrow R = 1 \cdot \frac{12 \times 10^{-2}}{(50 \times 10^{-6}) \cdot (20 \times 10^{-6})}$ $\Rightarrow R = 12 \times 10^7 \Omega$

$R = \frac{V}{I} \Rightarrow V = R \cdot I \Rightarrow V = 1200$ V

Regarding the dissipated heat, we will have that:

this is $\leftarrow Q = R I^2 \Delta t$ $\Rightarrow Q = 12 \times 10^7 \cdot (10 \times 10^{-6})^2 \cdot \Delta t$ $\Rightarrow$
 $\xrightarrow{\text{time interval}}$

due to the joule effect the heat dissipated $\Rightarrow Q = 0,012$ J

dissipated heat

If we consider a time interval of 1 second, we have that the heat dissipated is 0,012 J.

② Determine the electro-osmotic flow rate.

Explain! Consider only flow contribution from electro-osmotic flow.

The electro-osmosis is a non-equilibrium effect, where a liquid is brought to move relative to a charged surface by an applied external potential gradient $\nabla\phi_{\text{ext}}$.



that is, basically it is an effect in which the liquid will move due to the action of an electric field E , caused by the electrodes.

From the graph given in the statement we know that the electro-osmosis mobility, for the neutral buffer solution (pH=7), is $\mu_{eo} = 3,5 \times 10^{-4} \text{ cm}^2 / (\text{V} \cdot \text{s})$
 $\Rightarrow \mu_{eo} = 3,5 \times 10^{-8} \text{ m}^2 / (\text{V} \cdot \text{s})$

The electric field, which causes this effect, will be given by:

$$\underbrace{V = E \cdot dl}_{\text{from Maxwell's equations}} \Rightarrow E = \frac{V}{dl} \Rightarrow E = \frac{1200}{12 \times 10^{-2}} \Rightarrow E = 10000 \text{ V/m}$$

From here, we will now calculate the EO velocity, which is given by:

$$v_{eo} = \frac{E}{\eta} \quad \longrightarrow \quad \text{but know } \mu_{eo} = \frac{E}{\eta},$$

$$\begin{aligned} \text{we're going to have: } v_{eo} &= \mu_{eo} E \Rightarrow \\ \Rightarrow v_{eo} &= 3,5 \times 10^{-8} \cdot 10000 \Rightarrow v_{eo} = 3,5 \times 10^{-4} \text{ m/s} \end{aligned}$$

So, from here, we will have that the electro-osmotic flow rate is:

$$\begin{aligned} Q_{eo} &= v_{eo} \cdot \underbrace{A}_{\text{channel area}} \Rightarrow Q_{eo} = 3,5 \times 10^{-4} \text{ m/s} \Rightarrow \\ &\Rightarrow Q_{eo} = 3,5 \times 10^{-4} \cdot (50 \times 10^{-6}) \cdot (20 \times 10^{-6}) \Rightarrow \\ &\Rightarrow Q_{eo} = 3,5 \times 10^{-13} \text{ m}^3/\text{s} \end{aligned}$$

③ What is the pressure this electro-osmotic flow rate is causing? Why? Explain how you calculated it!

In this exercise 3, we want to know the pressure that the electro-osmotic flow rate, which we calculated in exercise 2, is causing.

Now, for this we have to apply the Hagen Poiseuille law, which basically says that:

$\Delta p = R_{hyd} Q_{eo}$
this is the pressure that we want to know $\downarrow$?? $\rightarrow$ this is what we got in the previous exercise

Therefore, in order to know the pressure, we have to find out this resistance of the fluid R_{hyd} . So, we are going to start by defining the fluid for water.

From that, the R_{hyd} value is going to be:

$$R_{hyd} \approx \frac{12}{1 - 0,63 \times \left(\frac{20 \times 10^{-6}}{50 \times 10^{-6}} \right)} \cdot 0,001002 \cdot 12 \times 10^{-2} \frac{1}{(20 \times 10^{-6})^3 (50 \times 10^{-6})} \quad (\text{E})$$

$$\text{E) } R_{hyd} \approx 4,822 \times 10^{15} \text{ Pa s/m}^3$$

So, the pressure this electro-osmotic flow rate is causing is:

$$\Delta p = 4,822 \times 10^{15} \cdot 3,5 \times 10^{-13} \quad (\text{E}) \quad \Delta p = 1687,86 \text{ Pa}$$

this pressure is due to the fact that we have a movement of this electro-osmotic flow that encounters resistance in the fluid.

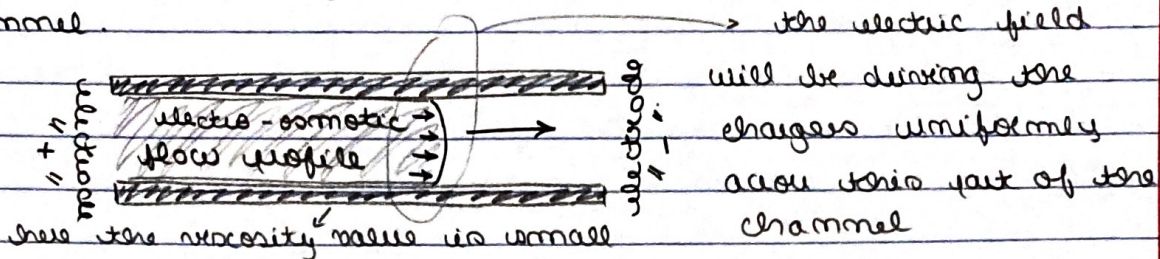
④ Depict flow and flow profiles at both sides of the "-" electrode! What is the flow direction?

We will start by analyzing the flow and the flow profiles before the "-" electrode:

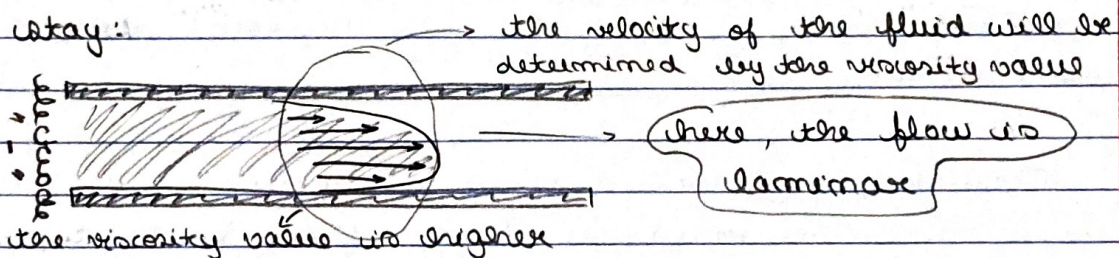
$\hookrightarrow$ Here the fluid is being moved due to the action of an electric field, which will imply that the flow

$\rightarrow$

profile is almost flat. Well, this happens because the force propelling the fluid is distributed along the channel.



Now looking at the part that is after the "+" electrode, we know that there is no electric field applied. So, in this one it will be the pressure, that we will apply, that will cause flow. Here it will stay:



here the viscosity value is higher

Now we will focus on the flow direction:

→ We know that when a fluid comes in contact with a dielectric wall (in this case is the electrode), there is going to be a charge separation near the solid-liquid interface. So, the surface charge will attract the opposite charge ions, creating an that forms a layer of these ions. Therefore, when we applied the potential between the electrodes, this generates an electrical body force on these ions, making them move to the opposite charged electrode.

↓

in this case, in the first part the direction of the flow is from "+" to "-" and in the second part is from "-" to the end of the channel.

⑤ Calculate total flow rate from one side of the channel to another.

From the sliders, we know that:

$$Q = \Delta p \cdot \frac{d^3 \omega}{12 \eta (L-x)} \left[1 - 0,630 \frac{d}{\omega} \right]$$

$d = 20 \mu m$
 $\omega = 50 \mu m$
 $L = 12 cm$
 $x = 20 cm$

this is
the flow rate
after the " - "
electrode

→ we don't know the viscosity
value η of the fluid. So, we will
consider that the fluid is water. Therefore,
at $T = 20^\circ C$, we will have:

$$\eta = 1,002 cP = 0,001002 P$$

(it was used on exercise 3)

From there, all we need is to calculate the Δp after
the " - " electrode. Since from the " - " electrode, the
only thing that changes is the fact that there is no
longer a potential acting, so we can consider that
 $\Delta p = 1687,86 Pa$.

Therefore, we will have:

$$Q = 1687,86 \cdot \frac{(20 \times 10^{-6})^3 \cdot (50 \times 10^{-6})}{12 \cdot 0,001002 \cdot (20 - 12) \times 10^{-2}} \left[1 - 0,630 \frac{20 \times 10^{-6}}{50 \times 10^{-6}} \right] \quad (\Rightarrow)$$

$$\Rightarrow Q = 1687,86 \cdot 4,4583 \times 10^{-16} \cdot 0,748 \quad (\Rightarrow)$$

$$\Rightarrow Q = 5,2499 \times 10^{-13} \text{ m}^3/s$$

as we said above, this is only the flow rate
after the " - " electrode.

So, in conclusion, the total flow rate from one
side of the channel to another is:

$$Q_T = Q_{e0} + Q \quad (\Rightarrow) \quad Q_T = 3,5 \times 10^{-13} + 5,2499 \times 10^{-13} \quad (\Rightarrow)$$

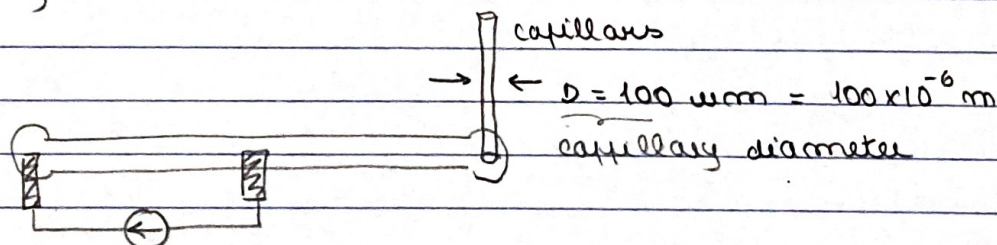
$$\Rightarrow Q_T = 8,75 \times 10^{-13} \text{ m}^3/s$$

⑥ Now at the end of the tube down to "-" electrode, there will be a vertical glass capillary connected to the channel outlet. Capillary diameter is $D = 100 \mu\text{m}$, contact angle of water and glass is $\theta = 55^\circ$, surface tension of the water is 1 mJ/m^2 . What is the height of the water in the capillary? (Electrical current is applied same way as previously).

* Further assumptions: do not assume that electrical current is causing any gas formation on the electrode.

Useful hint: Kirchhoff rules and superimpositions would be useful ideas how to tackle these flows (remember the analogy with electrical circuits!)

In this last exercise, we basically have the following situation:



And we want to know the height of the water in the capillary, which is given by:

$$H = \frac{2\gamma \cos(\theta)}{\rho \cdot g \cdot a}$$

fluid density (water $\rightarrow \rho = 997 \text{ kg/m}^3$)
 earth gravity
 capillary radius ($a = \frac{D}{2}$)
 contact angle of water and glass ($\theta = 55^\circ$)
 surface tension of the water ($\gamma = 1 \text{ mJ/m}^2$)
 $\Leftrightarrow \gamma = 1 \times 10^{-3} \text{ J/m}^2$

So,

$$H = \frac{2 \cdot 1 \times 10^{-3}}{997 \cdot 9.81 \cdot 100 \cdot 10^{-6} \cdot \frac{1}{2}} \cdot \cos(55^\circ) \quad (\Leftrightarrow)$$

$$\Leftrightarrow H = 0.002346 \text{ m} \quad \Leftrightarrow H = 2.346 \text{ mm}$$